

SMART BIKE SHARING OPERATIONAL EFFICIENCY & OPTIMIZATION

Objective:

The objective of this project is to analyze smart city bike-sharing data to evaluate system performance in terms of bike availability, station efficiency, and operational consistency across different cities and countries.

The project aims to identify high-performing and underperforming stations, understand geographical distribution patterns, analyze inventory mismatches, and provide data-driven recommendations to optimize bike redistribution and incentive programs.

Dataset Description:

The dataset consists of a station-level fact table containing real-time inventory data, along with supporting dimension tables for Stations and Bikes.

The fact table includes attributes like Station ID, Station Name, Address, City, Country, Status (Open/Closed), Banking options, Bonus eligibility, Available Bikes, Available Bike Slots, Bike Stands, and Timestamps (Date, Hour, Minutes).

The dimension tables provide additional descriptive information related to station locations and bike stand capacities, enabling detailed analysis of operational efficiency, reward category effectiveness, and inventory accuracy across various European cities.

Fact Table (Smart_Bike_Fact) – Column Description:

Column Name	Description
Station_ID	A unique identifier assigned to each bike station. Used to track and reference individual station records.
Station_Name	The name of the bike station for easy identification.
Address	The physical address of the station. Cleaned to remove special characters for consistency.
City	The city where the station is located (derived from Contract Name).
Country	The country where the station is located (derived from Latitude/Longitude).
Status	Operational status of the station (Open or Closed).
Banking	Indicates payment method availability (App-Only vs. Payment Terminal).
Contract_Name	The name of the contract or operator managing the station.

Available_Bikes	The current number of bikes available for rent at the station.
Available_Bike_Slots	The current number of empty slots available for bike returns.
Bike_Stands	The total capacity (stands) of the station.
Date	The date of the inventory record.
Hour	The hour of the day the record was captured.
Minutes	The minute of the hour the record was captured.
Bonus	Indicates if the station is in an Incentive Zone or Standard Zone.
Reward_Category	Calculated category indicating the free ride incentive based on slot availability.
Inventory_Record	Calculated status validating if the inventory count matches capacity (Correct, Mismatch, Over Capacity).

Station Dimension Table (Station_Dim) – Column Description:

Column Name	Description
Station_ID	A unique identifier assigned to each station.
Station_Name	The cleaned name of the station.
Address	The standardized physical address.
City	The city location.
Country	The country location.
Status	Current operational status.
Position	Latitude and Longitude coordinates for mapping.
Banking	Payment terminal availability status.
Contract_Name	Operator contract name.

Bike Dimension Table (Bike_Dim) – Column Description:

Column Name	Description
Bike_Index_Key	A unique primary key created for bike stand tracking.
Station_ID	Linked identifier to associate bike stands with specific stations.
Bike_Stands	Total capacity of bike stands at the station.

Data Import Process:

The dataset was imported into the analysis environment using **Power BI Desktop**. The following steps were followed to load the data:

1. Navigated to the **Home** tab in Power BI Desktop.
2. Selected **Get Data** and chose **From Excel Workbook**.
3. Browsed and selected the **Bike-Stations-Sharing-Data** file.
4. Chose the required worksheet from the workbook.
5. Clicked **Transform Data**, and the necessary data cleaning steps were performed in **Power Query** before loading the data into the model for further analysis.

Data Cleaning & Transformation:

- **Station ID Standardized:**
The “Number” column was confirmed as the unique station identifier and renamed to Station_ID. Duplicate records with conflicting names were removed to prevent double-counting of bikes and slots.
- **Station Name Standardized:**
Unreadable Chinese and French characters were removed using a Custom Column. The station number was removed from the name text. Clean, Trim, and Proper functions were applied to ensure consistent text formatting.
- **Address Standardized:**
Unreadable characters were removed using the formula `Text.Select([address],{"A".. "Z", "a".. "z", "", "-"})`. Missing addresses were imputed using city names based on position. Special characters like “:” and “()” were standardized as hyphens.
- **Position Standardized:**
The Position column was split into Latitude and Longitude fields to enable map visualizations. Clean and Trim transformations were applied to remove unwanted spaces.
- **Banking Standardized:**
Boolean values were standardized for clarity: True → Station has a payment terminal; False → App-only payment.
- **Status Standardized:**
Values were trimmed and capitalized consistently: OPEN → Operational; CLOSED → Not operational.

- **Contract Name Standardized:**

Extra spaces and inconsistent capitalization were removed (e.g., "maribor" → "Maribor").

- **Bonus Standardized:**

Boolean values were replaced with descriptive categories:

True → Incentive Zone; False → Standard Zone.

- **Numeric Columns (Bikes, Slots, Stands):**

Clean and Trim applied. Zero values were retained as they represent critical operational states (empty/full) and were not replaced with averages.

- **Date & Time Split:**

The Date column was split into Date, Hour, and Minutes to enable hourly trend analysis.

- **Reward Category (Calculated):**

A custom column was created to incentivize returns based on slot availability.

Formula :

```
Reward Category = SWITCH(TRUE(),  
Smart_Bike_Fact[Available Bike Slots]>=30 && Smart_Bike_Fact[Available Bike  
Slots]<=40, "+15 Min Free Next Ride",  
Smart_Bike_Fact[Available Bike Slots]>=20 && Smart_Bike_Fact[Available Bike  
Slots]<=29, "+10 Min Free Next Ride",  
Smart_Bike_Fact[Available Bike Slots]>=10 && Smart_Bike_Fact[Available Bike  
Slots]<=19, "+5 Min Free Next Ride",  
Smart_Bike_Fact[Available Bike Slots]>=5 && Smart_Bike_Fact[Available Bike Slots]<=9,  
"+3 Min Free Next Ride",  
"No Bonus")
```

- **Inventory Record (DAX Calculation):**

A validation column was created to check data integrity.

Formula :

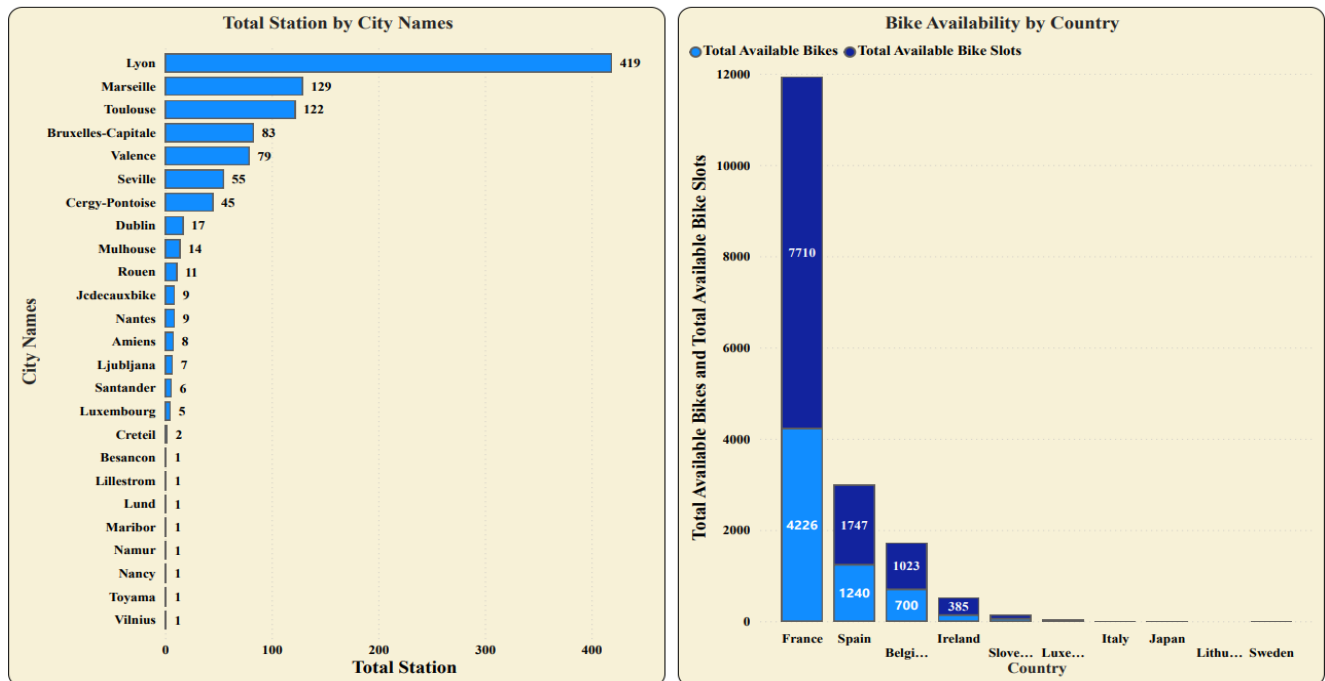
```
Inventory Record = IF(Smart_Bike_Fact[Available Bike Slots] + Smart_Bike_Fact[Available  
Bikes] > Smart_Bike_Fact[Bike Stands],  
IF(RELATED(Station_Dim[Status])="Closed","Closed Station","Over Capacity"),  
IF(Smart_Bike_Fact[Available Bike Slots] + Smart_Bike_Fact[Available Bikes] =  
Smart_Bike_Fact[Bike Stands],"Correct","Mismatch"))
```

- **Data Modeling:**

Relationships were established between Station_Dim and Fact Table (via Station_ID) and Bike_Dim and Fact Table (via Bike_Index_Key).

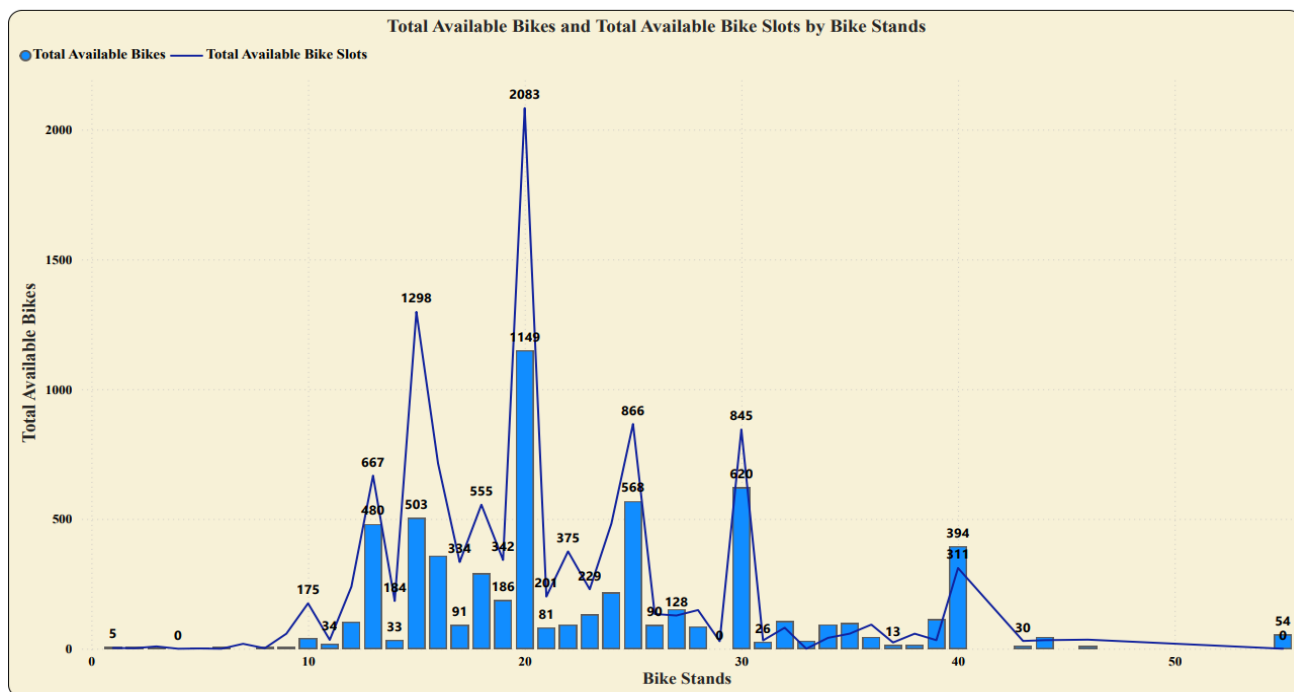
Visualization and Insights

Total Stations by City & Bike Availability by Country



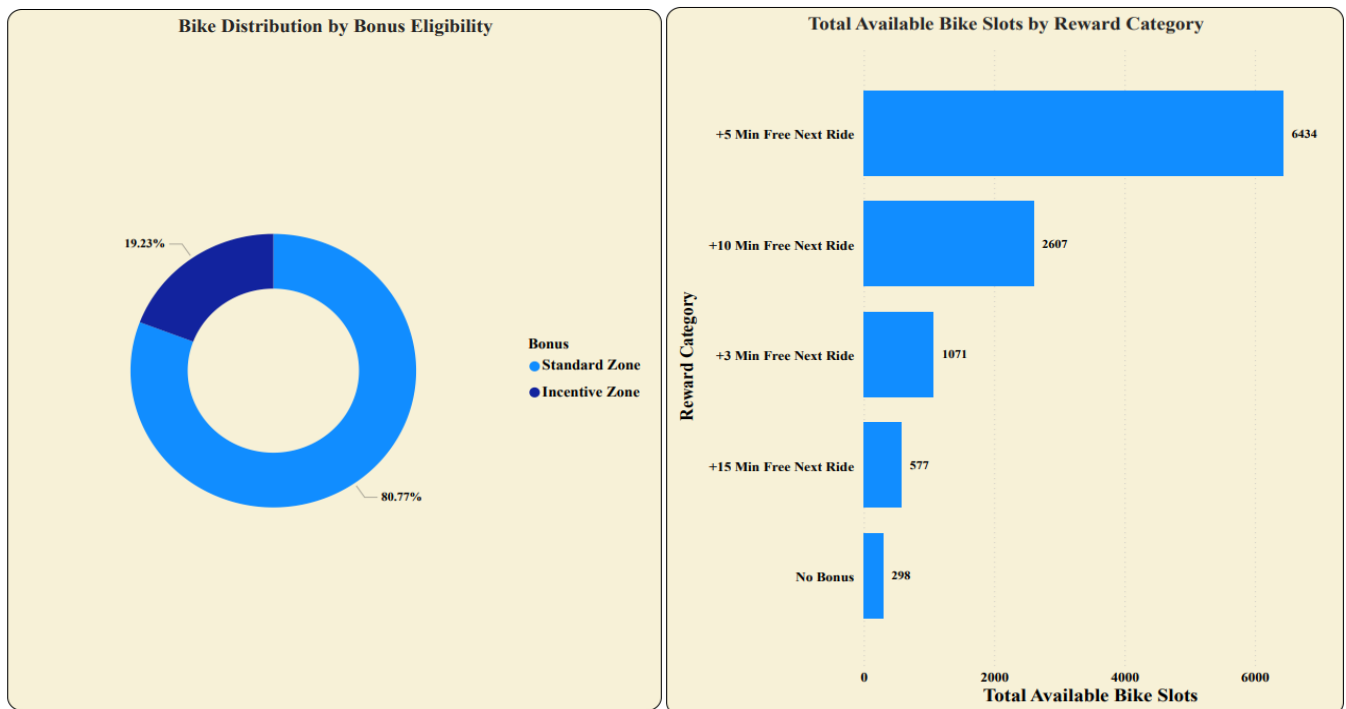
- Lyon has the highest number of stations (419), followed by Marseille (129) and Toulouse (122). France dominates the system with 4,226 available bikes and 7,710 slots. Spain and Belgium follow with smaller but significant networks.
- The heavy concentration in Lyon suggests a mature market, while cities with <10 stations (e.g., Lund, Namur) may be in pilot phases or facing adoption barriers. The disparity between bike counts and slot counts in France indicates potential over-provisioning of infrastructure relative to current bike supply.
- If current expansion trends continue (steady growth from 2022–2025), France will likely maintain its dominance, while smaller markets may require targeted investment to reach critical mass.
- Prioritize maintenance and bike redistribution resources in high-density hubs like Lyon to prevent bottlenecks. For smaller cities, conduct feasibility studies before expanding station counts to ensure demand matches supply.

Bike Availability by Hour & Station Status



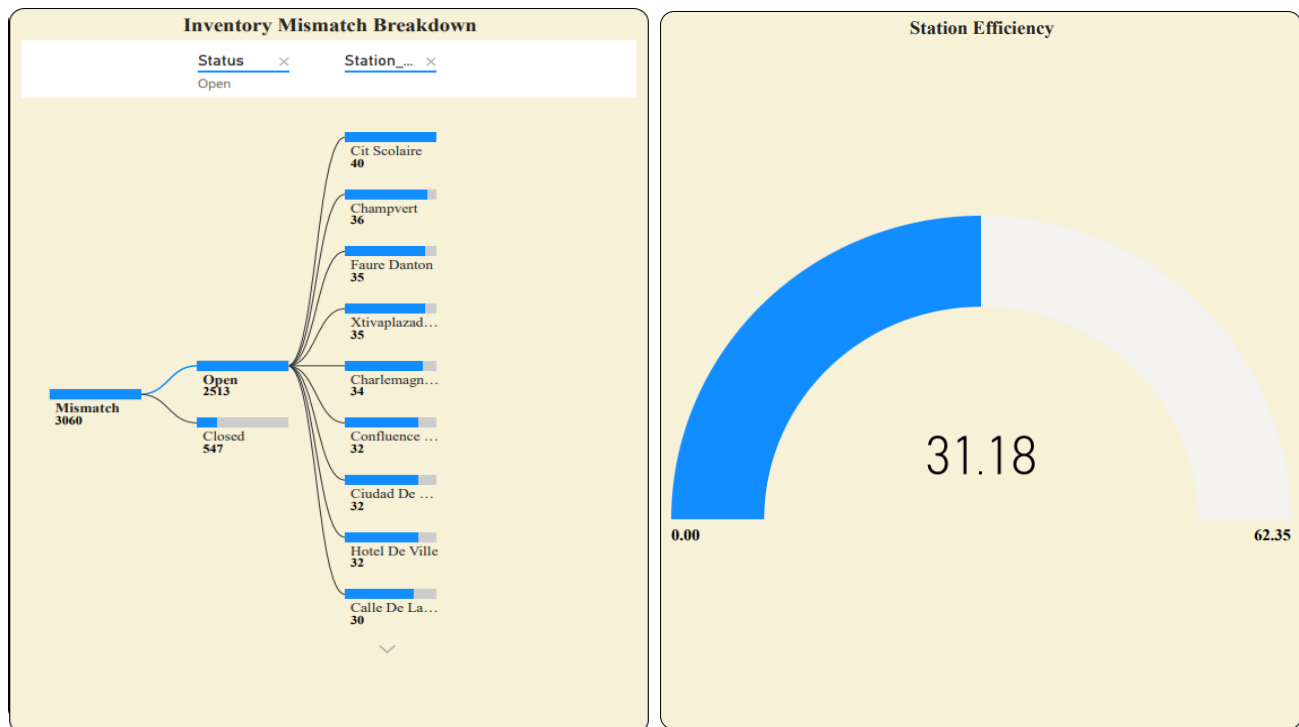
- Bike availability fluctuates throughout the day, with lower availability during peak commuting hours. 913 stations are Open, while 115 are Closed.
- The dip in available bikes during specific hours confirms commuter-driven demand cycles. The 11.2% closure rate may be due to scheduled maintenance or technical failures, impacting local accessibility.
- Peak demand windows will likely persist during morning (8–9 AM) and evening (5–6 PM) rush hours.
- Closure rates may spike during extreme weather or holiday periods if not proactively managed.
- Implement dynamic staffing for bike redistribution just before peak hours.
- Monitor closed stations closely and set SLAs for maintenance teams to reduce downtime below 5%.

Reward Category & Bonus Zone Distribution



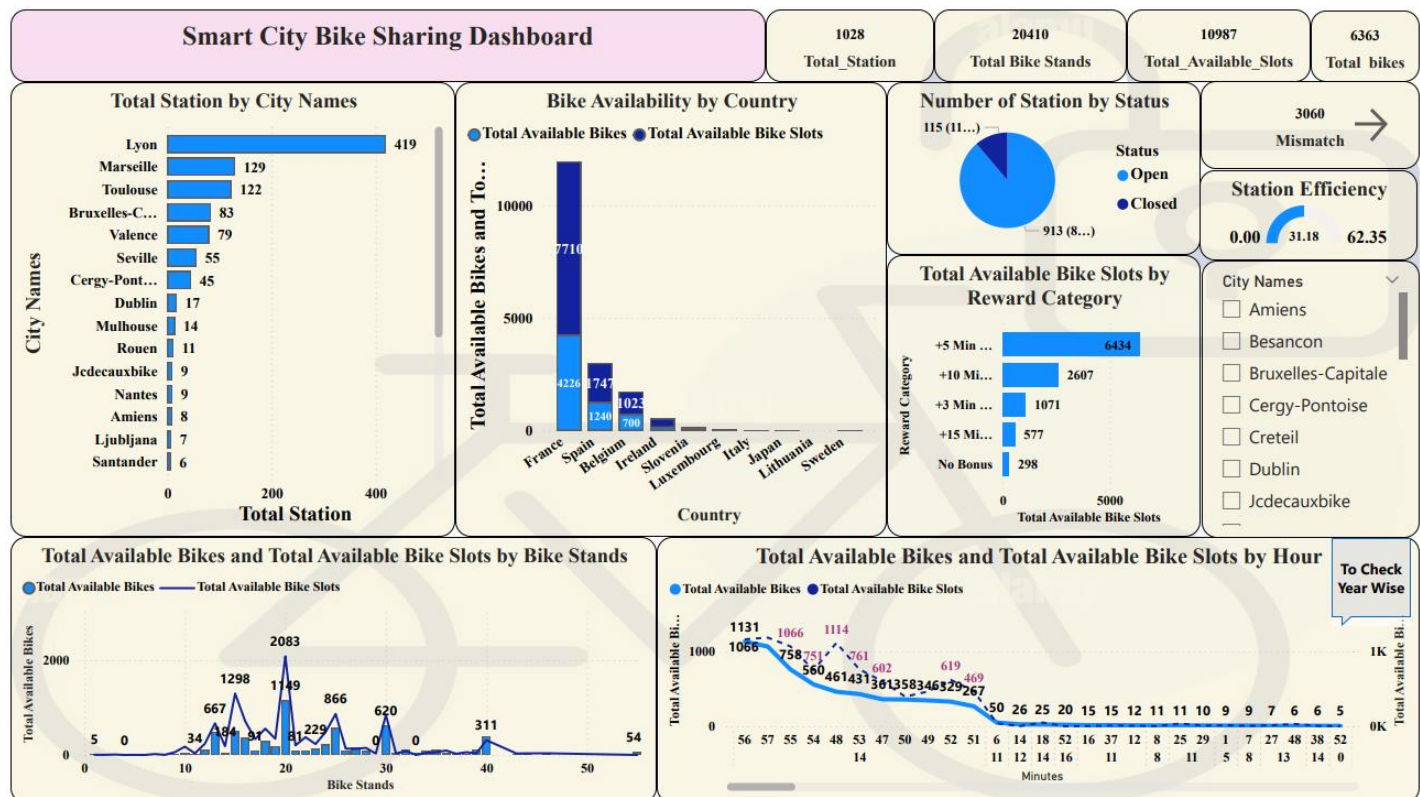
- 80.77% of stations are in Standard Zones, while 19.23% are Incentive Zones. The "+5 Min Free Next Ride" is the most common reward category (6,434 slots), followed by "+10 Min" (2,607 slots).
- The prevalence of the "+5 Min" reward suggests that many stations frequently hit the 10–19 slot availability threshold, indicating moderate imbalance.
- The lower frequency of higher rewards (+15 min) implies fewer stations reach critical vacancy levels.
- As the network grows, the frequency of high-value incentives (+15 min) may increase if redistribution strategies are not optimized, leading to higher operational costs.
- Analyze the geographic clusters of Incentive Zones to identify chronic "sink" or "source" stations.
- Adjust the reward thresholds dynamically based on real-time data to optimize cost-efficiency of the incentive program.

Inventory Mismatch & Station Efficiency



- There are 3,060 inventory mismatches, with 2,513 occurring in Open stations. Top mismatch locations include "Cit Scolaire" and "Champvert".
- Average station efficiency is 31.18%.
- High mismatches in Open stations indicate sensor errors, data transmission lag, or theft/vandalism rather than simple closure issues.
- Low average efficiency (31%) suggests that stations are often either too full or too empty, failing to utilize their mid-range capacity effectively.
- Without intervention, mismatch rates could erode user trust, leading to decreased ridership.
- Stations with recurring mismatches are likely to experience higher maintenance costs.
- Conduct immediate operational audits on top mismatch stations (e.g., Cit Scolaire).
- Install IoT sensors with higher reliability or implement manual verification protocols.
- Redesign the redistribution algorithm to target stations with low efficiency scores first.

Conclusion



- The bike-sharing dashboard provides a comprehensive view of station distribution, bike availability, and operational health across major European cities.
- France, particularly Lyon, leads the network in scale and infrastructure, demonstrating strong adoption. However, the high concentration also presents risks of localized congestion.
- While the majority of stations are operational, the significant volume of **inventory mismatches (3,060)** highlights critical data integrity and tracking issues that need immediate attention.
- The current incentive program is active but skewed towards lower-tier rewards, suggesting moderate success in balancing supply.
- Seasonal and hourly trends confirm commuter-based usage patterns, necessitating proactive redistribution strategies during peak times.
- By addressing inventory discrepancies, optimizing incentive zones, and improving maintenance response times, the system can significantly enhance station efficiency and user satisfaction.
- Overall, the insights derived enable informed decision-making to expand the network sustainably, reduce operational costs, and ensure reliable service for smart city commuters.